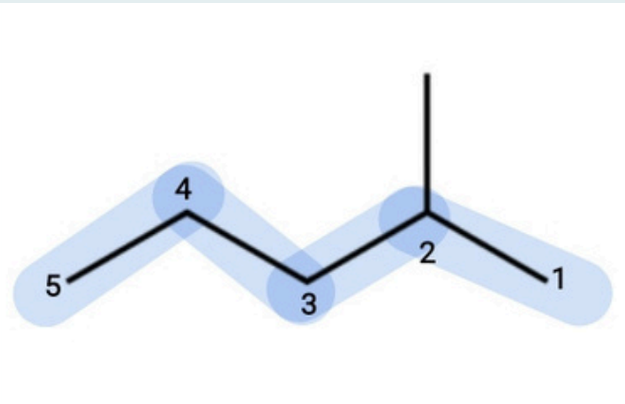
PART A

STRUCTURE → NAME

CARD A01 — STRUCTURE → NAME

Level: Foundation Molecular formula: C_6H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 5 C → Pentane. Other one 5-C paths exist, but each carries one methyl group.

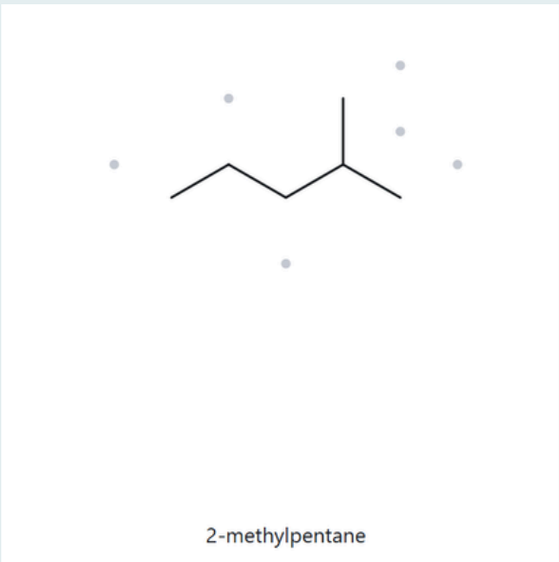
Numbering: left → right gives 4; right → left gives 2. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2 → 2-methyl.

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



Returned name: 2-methylpentane
✓ Match

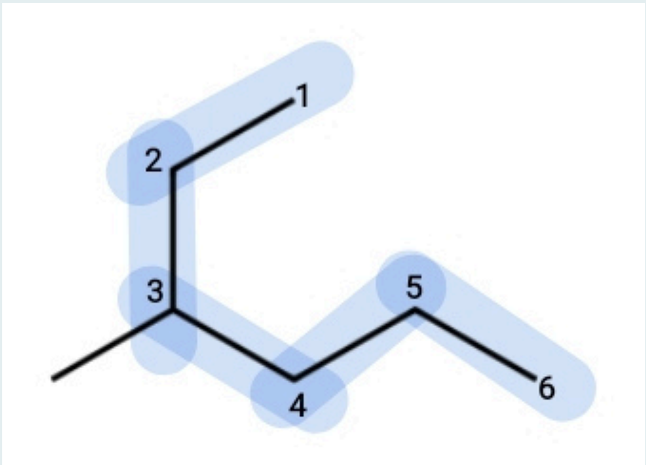
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A02 — STRUCTURE → NAME

Level: Foundation Molecular formula: C_7H_{16}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C → Hexane. Several 6-C paths exist, but each carries one methyl group.

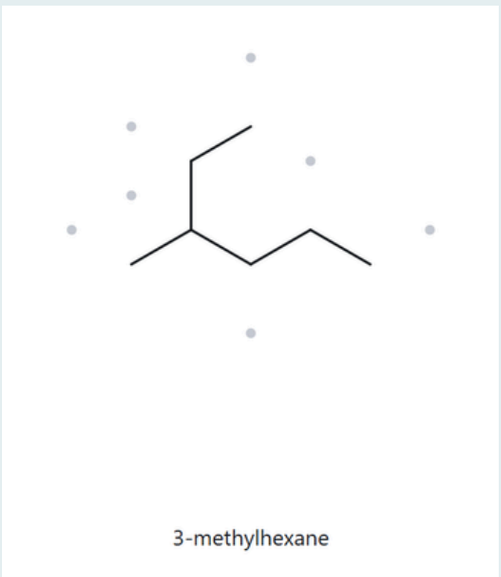
Numbering: left → right gives 3; right → left gives 4. First point of difference: 3 < 4, so number from the left.

Substituents: methyl on C3 → 3-methyl.

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



Returned name: 3-methylhexane
✓ Match

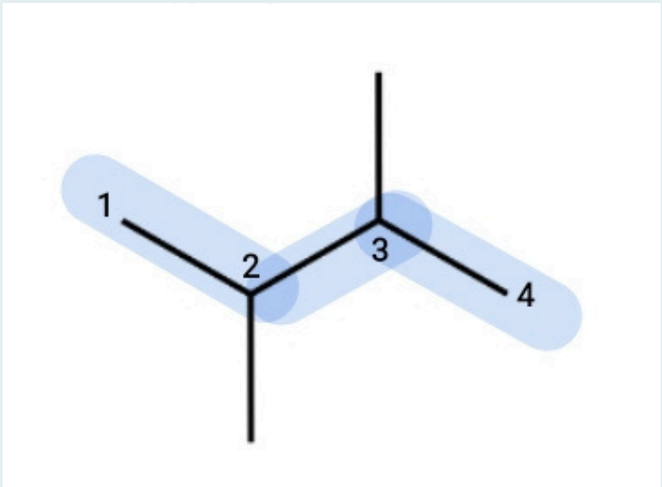
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A03 — STRUCTURE → NAME

Level: Foundation Molecular formula: C_6H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries two methyl groups.

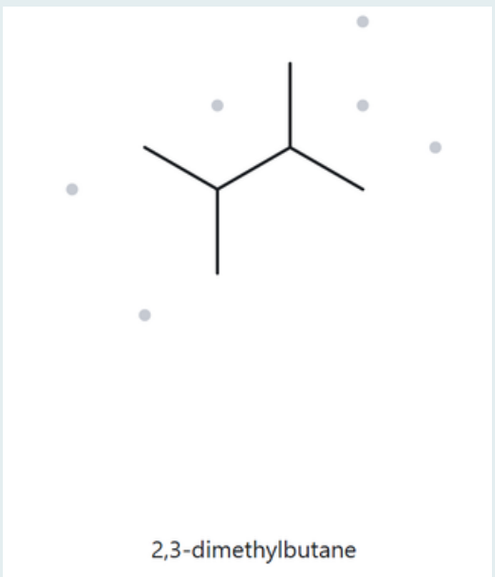
Numbering: left → right gives 2,3; right → left gives 3,2. First point of difference: 2 < 3, so number from the left.

Substituents: methyl on C2 and C3 → 2,3-dimethyl.

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



Returned name: 2,3-dimethylbutane
✓ Match

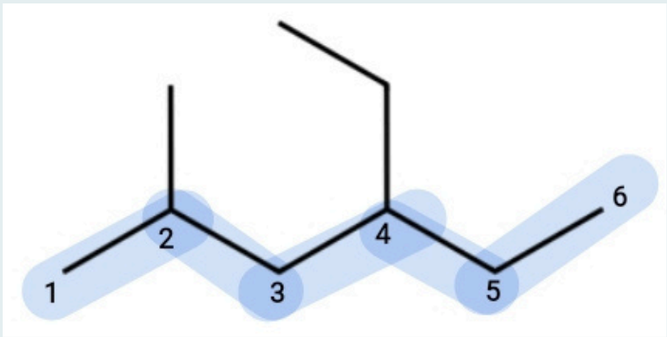
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A04 — STRUCTURE → NAME

Level: Developing Molecular formula: C_9H_{20}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C → hexane. Several 6-C paths exist, but each carries one methyl group and one ethyl group.

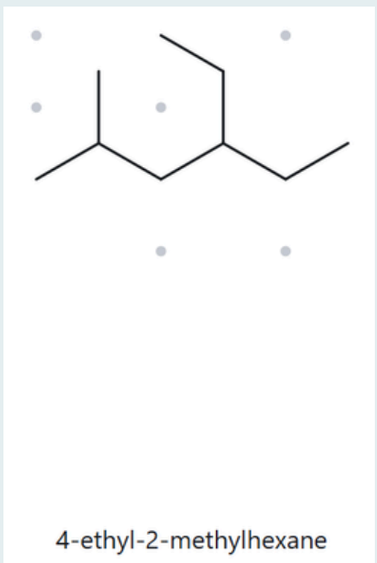
Numbering: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.

Substituents: methyl on C2 and ethyl on C4 → 4-ethyl-2-methyl

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



Returned name: 4-ethyl-2-methylhexane
✓ Match

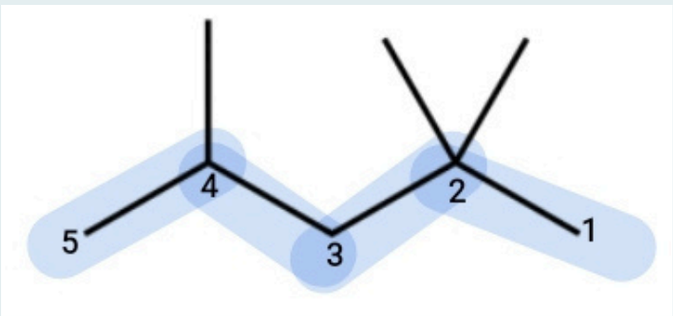
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ alphabetical order | ✓ punctuation | ✓ formula counted

CARD A05 — STRUCTURE → NAME

Level: Developing Molecular formula: C_8H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 5 C → hexane. Several 5-C paths exist, but each carries three methyl groups.

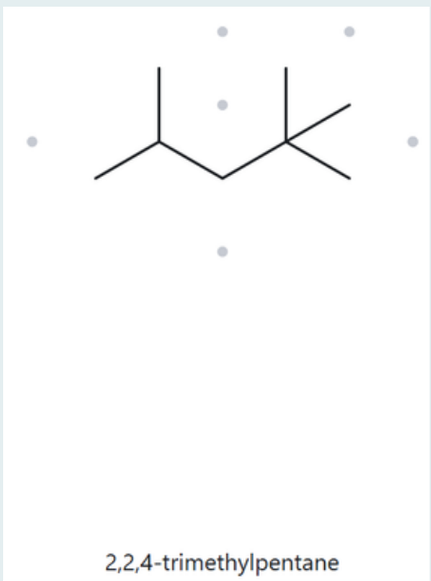
Numbering: left → right gives 2,4,4; right → left gives 2,2,4. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2 and C4 → 2,2,4-trimethyl.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



Returned name: 2,2,4-trimethylpentane

✓ **Match**

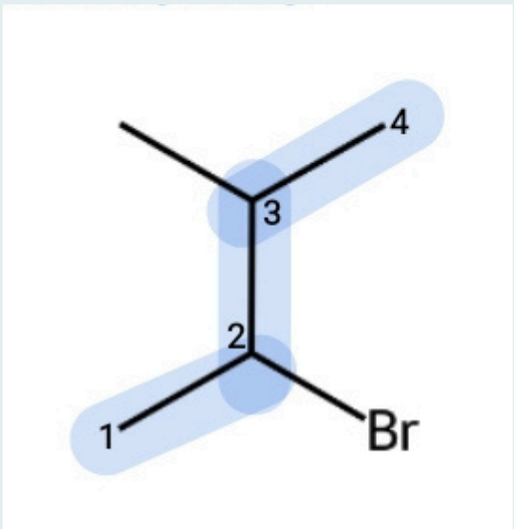
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A06 — STRUCTURE → NAME

Level: Developing Molecular formula: $C_5H_{11}Br$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries bromine and one methyl group.

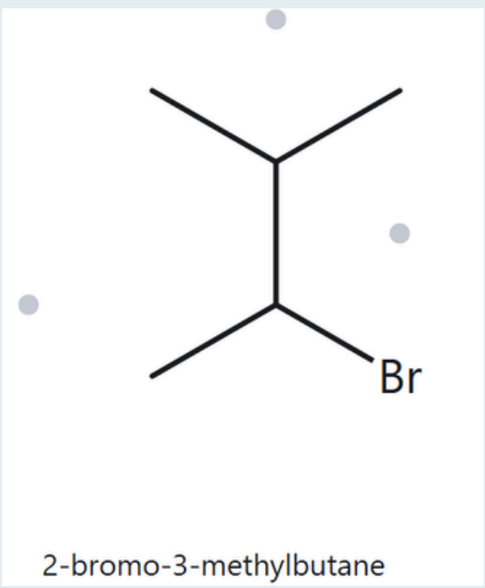
Numbering: left → right gives 2,3; right → left gives 2,3. Same point, either direction, so use the alphabetical tie-breaker.

Substituents: bromo on C2 and methyl on C3 → 2-bromo, 3-methyl

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



Returned name: 2-bromo-3-methylbutane

✓ **Match**

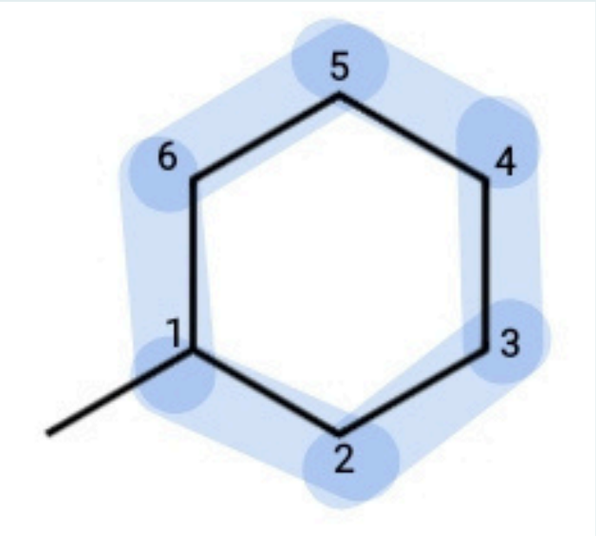
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ alphabetical order | ✓ punctuation | ✓ formula counted

CARD A07 — STRUCTURE → NAME

Level: Developing Molecular formula: C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Ring present → ring is the parent → cyclohexane (6 C).

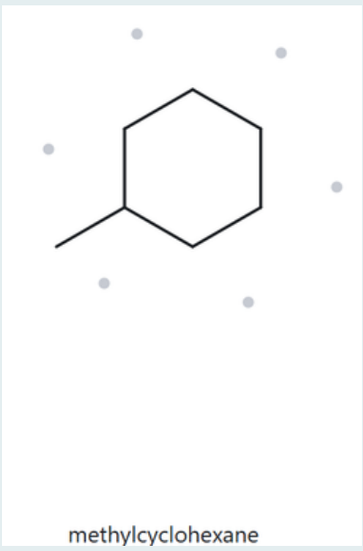
Numbering: Only one substituent is attached to the ring, so a locant is unnecessary.

Substituents: methyl on C1→ 1-methyl

3. MY NAME

methylcyclo hexane

4. THE ALKANE CHECK



Returned name: methylcyclohexane

✓ Match

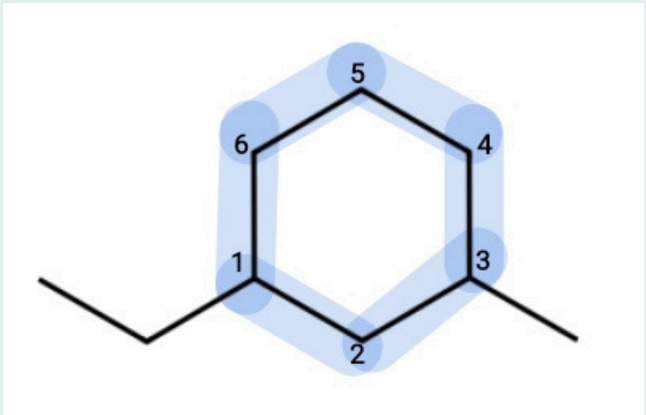
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A08 — STRUCTURE → NAME

Level: Challenge Molecular formula: C_9H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Ring present → ring is the parent → cyclohexane (6 C).

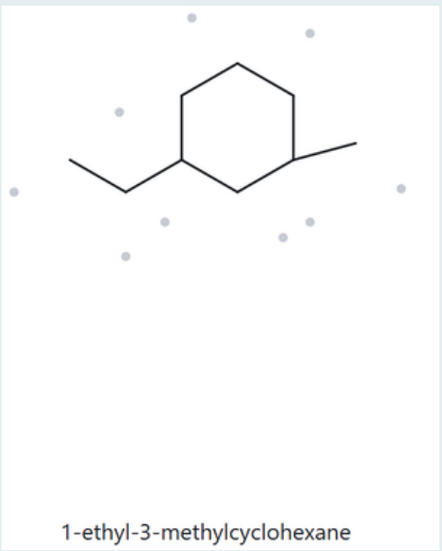
Numbering: Both directions give locants 1 and 3, so use the alphabetical tie-breaker.

Substituents: ethyl on C1 and methyl on C3 → 1-ethyl,3-methyl

3. MY NAME

1-ethyl-3 -methylcyclo hexane

4. THE ALKANE CHECK



Returned name: 1-ethyl-3-methylcyclohexane

✓ Match

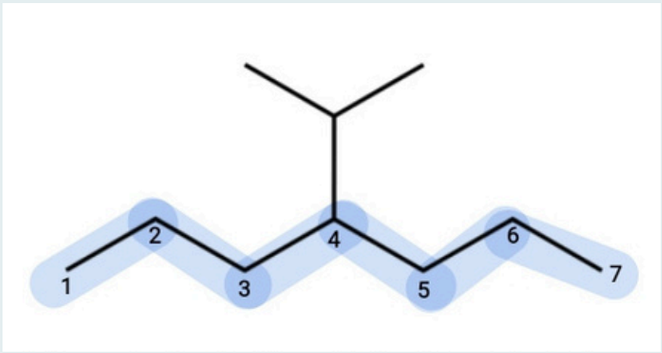
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ alphabetical order | ✓ punctuation | ✓ formula counted

CARD A09 — STRUCTURE → NAME

Level: Challenge Molecular formula: $C_{10}H_{22}$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 7 C → heptane. The longest chain runs along the bottom zigzag, and the upper branch is a three-carbon substituent attached through its middle carbon.

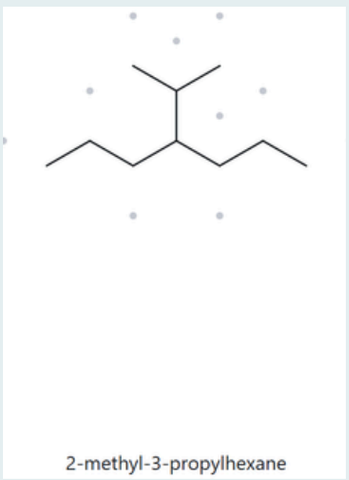
Numbering: Both directions give locant 4.

Substituents: isopropyl on C4 → 4-isopropyl

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



Returned name: 2-methyl-3-propylhexane

✗ Mismatch

Explanation: The longest continuous chain contains 7 carbon atoms. Therefore, the correct parent is heptane, giving **4-isopropylheptane.**

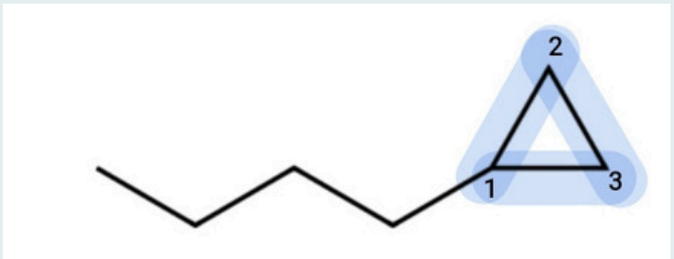
VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

CARD A10 — STRUCTURE → NAME

Level: Challenge Molecular formula: C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Ring present → ring is the parent → cyclopropane (3 C).

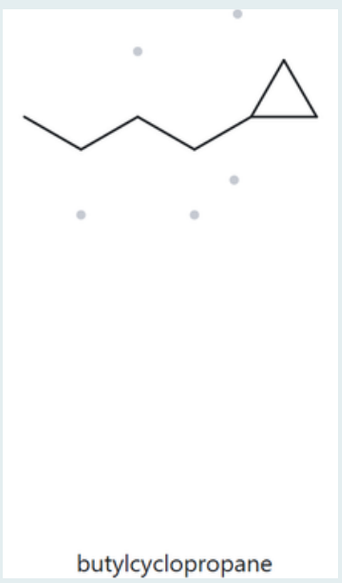
Numbering: A ring with one substituent needs no locant.

Substituents: butyl on C1 → butyl

3. MY NAME

butylcyclo propane

4. THE ALKANE CHECK



Returned name: butylcyclopropane

✓ **Match**

VERIFICATION STRIP

✓ longest chain/ring | ✓ lowest locants | ✓ punctuation | ✓ formula counted

PART B

NAME → STRUCTURE

CARD B01 — NAME → STRUCTURE

Level: Foundation Given name: 2-methylpropane

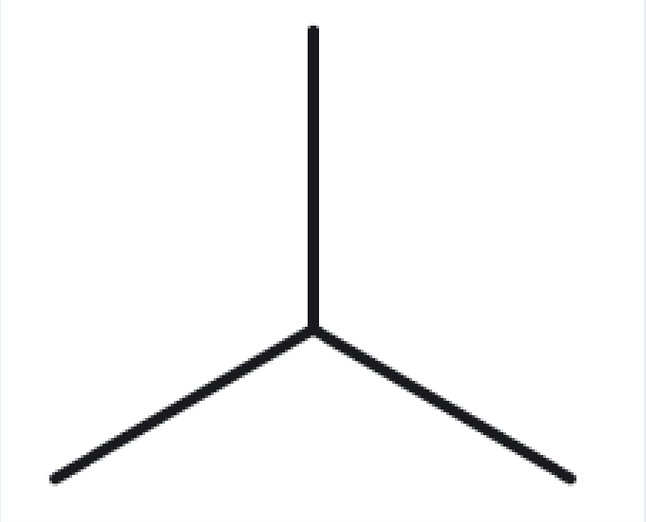
1. NAME DECODED

Parent: propane→ a 3-carbon chain

Substituents: methyl on C2

Expected formula: 3 + 1 = 4 C; acyclic alkane → **C_4H_{10}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_4H_{10}

Acyclic Alkane:

C_nH_{2n+2}= C_4H_{10}

C_4H_{10} ✓

4. BACK-CHECK

Returned by The Alkane:

2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B02 — NAME → STRUCTURE

Level: Foundation Given name: 3-ethylpentane

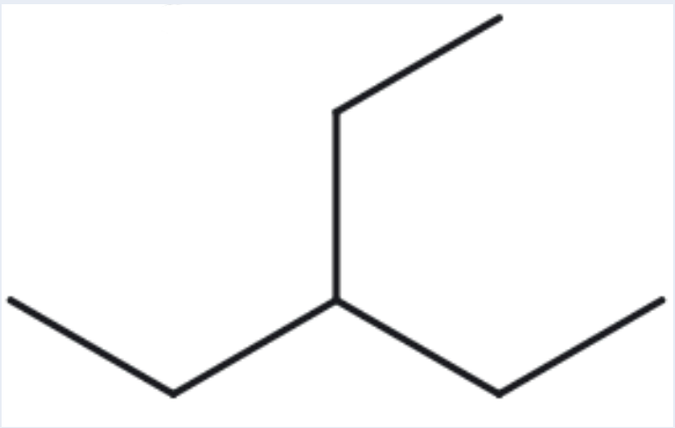
1. NAME DECODED

Parent: pentane→ a 5-carbon chain

Substituents: ethyl on C3

Expected formula: 5 + 2 = 7 C; acyclic alkane → **C_7H_{16}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:
C_7H_{16}

Acyclic Alkane:
$C_nH_{2n+2} = C_7H_{16}$

C_7H_{16} ✓

4. BACK-CHECK

Returned by The Alkane:
3-ethylpentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B03 — NAME → STRUCTURE

Level: Foundation Given name: 2,2-dimethylbutane

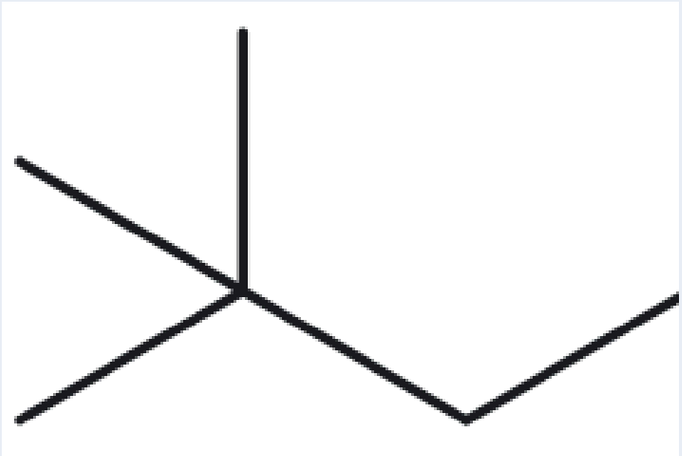
1. NAME DECODED

Parent: butane→ a 4-carbon chain

Substituents: methyl on C2

Expected formula: 4 + 2 = 6 C; acyclic alkane → **C_6H_{14}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_7H_{16}

Acyclic Alkane:

$C_nH_{2n+2} = C_6H_{14}$

C_6H_{14} ✓

4. BACK-CHECK

Returned by The Alkane:

2,2-dimethylbutane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B04 — NAME → STRUCTURE

Level: Developing Given name: 3-ethyl-2-methylpentane

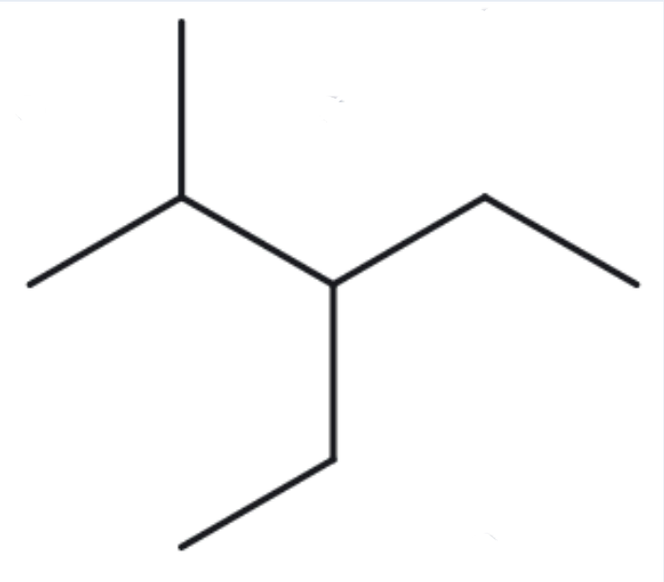
1. NAME DECODED

Parent: pentane → a 5-carbon chain

Substituents: ethyl on C3 ; methyl on C2

Expected formula: 5 + 2 + 1 = 8 C; acyclic alkane → **C_8H_{18}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_8H_{18}

Acyclic Alkane:

$C_nH_{2n+2} = C_8H_{18}$

C_8H_{18} ✓

4. BACK-CHECK

Returned by The Alkane:

3-ethyl-2-methylpentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B05 — NAME → STRUCTURE

Level: Developing Given name: 1-chloro-2-methylpropane

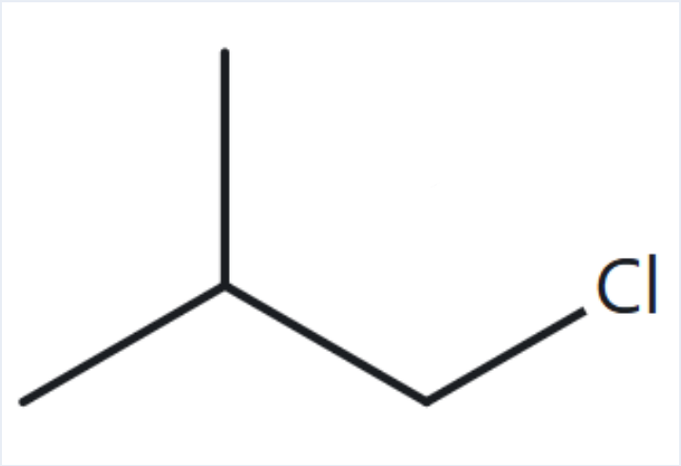
1. NAME DECODED

Parent: propane → a 3-carbon chain

Substituents: chloro on C1 ; methyl on C2

Expected formula: 3 + 1 = 4 C; acyclic alkane → C_4H_{10}; Cl replaces one H → **C_4H_9Cl**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:
C_4H_9Cl

Acyclic Alkane:
C_nH_{2n+2} = C_4H_{10}; minus one H for Cl =C_4H_9Cl

C_4H_9Cl ✓

4. BACK-CHECK

Returned by The Alkane:
1-chloro-2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B06 — NAME → STRUCTURE

Level: Developing Given name: 1,2-dimethylcyclopentane

1. NAME DECODED

Parent: cyclopentane→ a 5-carbon ring

Substituents: methyl on C1 ; methyl on C2

Expected formula: 5 + 2 = 7 C; one ring → **C_7H_{14}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_7H_{14}

One Ring:

$C_nH_{2n} = C_7H_{14}$

C_7H_{14} ✓

4. BACK-CHECK

Returned by The Alkane:

1,2-dimethylcyclopentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B07 — NAME → STRUCTURE

Level: Developing Given name: 4-ethyl-3,3-dimethylheptane

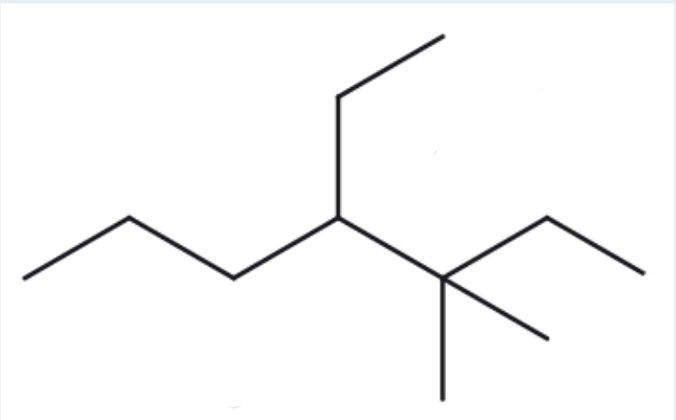
1. NAME DECODED

Parent: heptane→ a 7-carbon chain

Substituents: ethyl on C4; methyl on C3

Expected formula: 7 + 2 + 2 = 11 C; acyclic alkane → **$C_{11}H_{24}$**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

$C_{11}H_{24}$

Acyclic alkane:

$C_nH_{2n+2} = C_{11}H_{24}$

$C_{11}H_{24}$ ✓

4. BACK-CHECK

Returned by The Alkane:

4-ethyl-3,3-dimethylheptane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B08 — NAME → STRUCTURE

Level: Challenge Given name: tert-butylcyclohexane

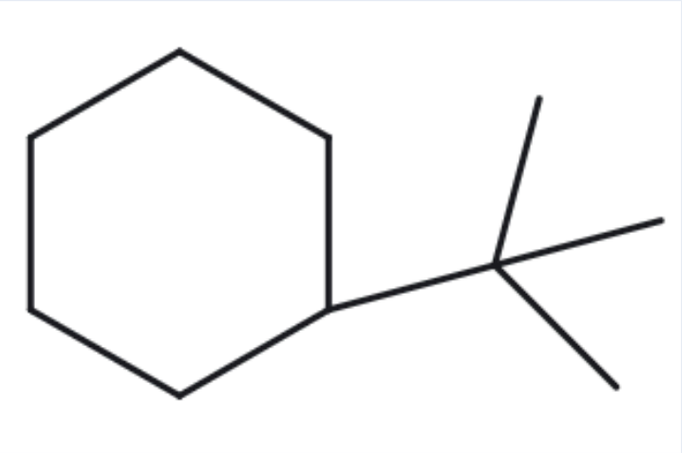
1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring

Substituents: tert-butyl on C1

Expected formula: 6 + 4 = 10 C; one ring → **$C_{10}H_{24}$**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

$C_{10}H_{24}$

One Ring:

$C_nH_{2n} = C_{10}H_{24}$

$C_{10}H_{24}$ ✓

4. BACK-CHECK

Returned by The Alkane:

(1,1-dimethylethyl)cyclohexane

✓ Same as the given name

Explanation: "tert-Butyl" is the common name for the substituent, while "1,1-dimethylethyl" is its systematic name. Both names describe the same substituent and therefore refer to the same compound. Both represent the same group:

$-C(CH_3)_3$

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B09 — NAME → STRUCTURE

Level: Challenge Given name: 3-ethyl-1,1-dimethylcyclohexane

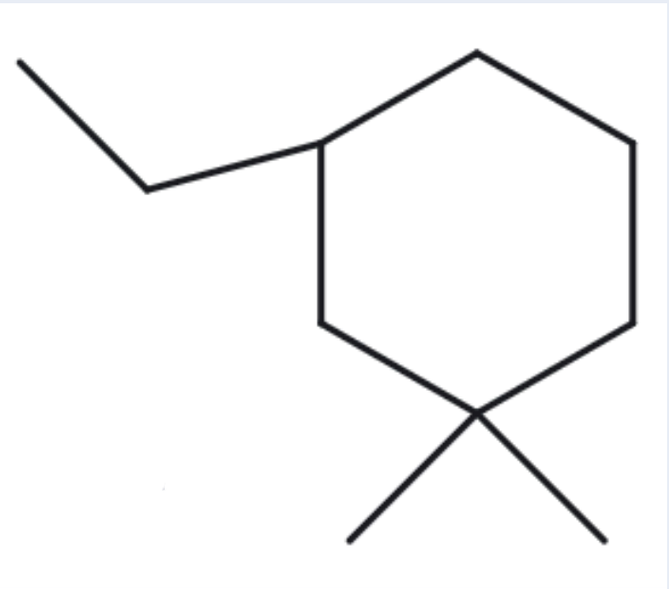
1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring

Substituents: ethyl on C3; methyl on C1

Expected formula: 6 + 2 + 2 = 10 C; one ring → **$C_{10}H_{20}$**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:
$C_{10}H_{20}$

One Ring:
$C_nH_{2n} = C_{10}H_{20}$

$C_{10}H_{20}$ ✓

4. BACK-CHECK

Returned by The Alkane:
3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check

CARD B010 — NAME → STRUCTURE

Level: Challenge Given name: 2-ethylbutane

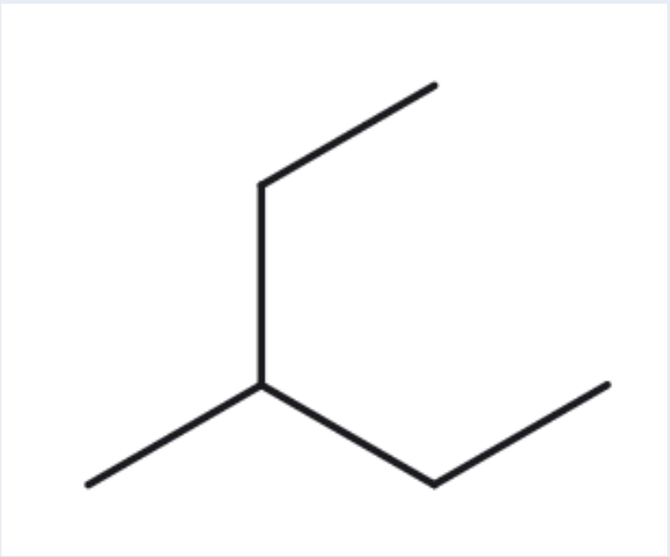
1. NAME DECODED

Parent: butane→ a 4-carbon chain

Substituents: ethyl on C2;

Expected formula: 4 + 2 = 6 C; acyclic alkane → **C_6H_{14}**

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_6H_{14}

Acyclic Alkane:

$C_nH_{2n+2} = C_6H_{14}$

C_6H_{14} ✓

4. BACK-CHECK

Returned by The Alkane:

3-methylpentane

✗ Given name is not the correct IUPAC name

Explanation: The given name, 2-ethylbutane, uses a four-carbon parent chain. However, the structure has a longer continuous chain of five carbon atoms, so the parent name must be pentane. The remaining carbon is a methyl substituent attached to carbon 3. Therefore, the correct IUPAC name is **3-methylpentane.**

VERIFICATION STRIP

✓ parent length matches | ✓ positions | ✓ formula | ✓ back-check